MIL-W-80040E
29 March 1989
SUPERSEDING
MIL-W-80040D
25 August 83

MILITARY SPECIFICATION

WELDING MACHINE, ARC, DC, TRANSFORMER-RECTIFIER, CONSTANT VOLTAGE, FOR MULTI-OPERATOR SYSTEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers constant voltage, direct current, transformer-rectifier, welding power sources for single or multi-operator systems requiring control stations with equipment and accessories specified herein. Control stations are covered by MIL-W-80042C.

1.2 Classification. The power source shall be of the following sizes. The size to be furnished shall be as specified (see 6.2).

Size	Rating, amperes (at 100% duty cycle)
500	500
750	750
1000	1000
1500	1500

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

(Unless otherwise Indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI 249.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA EW 1 - Electric Arc-Melding Apparatus. (DoD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code. (DoD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete power source shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The power source shall be new (not a prototype) and one of the manufacturer's current production models. Each power source shall include all components, parts, and features necessary to meet the performance requirements specified herein. Each power source shall be a single, free-standing, transportable, transformer-rectifier, direct current {DC), constant voltage power supply with associated instrumentation. The power source shall also be suitable to serve as a DC power source for multiple individual welding control

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grids within a multi-operator welding system. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dials. Graduations on dial and knob circumferences shall be permanently and legibly engraved, silkscreened, or etched on a nonglare background.

3.2.2 Reclaimed materials. The power source may contain reclaimed q aterlals to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the power source, Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to q ean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The welding power source and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the power source, whether or not specifically required herein, shall conform to all requirements of ANSI Z49.1 and OSHA 29 CFR 1910. If a conflict arises between ANSI 249.1 and OSHA 29 CFR 1910, OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer’s commercial power source supplied to the commercial market shall be provided for the point of operation and all other parts of the power source that present safety hazards.

3.2.5 Mercury restriction. The power source shall not contain mercury or q ercury compounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the power source.

3.2.7 Environmental protection. During the manufacture, operation, transportation, service, or storage of the power source, it shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect on the date of the contract at the point of installation.

3.2.8 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or power source. Replacement parts shall be domestically available.

3.3 Construction. The power source shall be constructed of parts which are new, without defects, and free of repairs. The power source shall withstand operation to its maximum rating and capacity without permanent distortion or failure.

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3.3.1 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently Installed.

3.3.2 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials, All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.3 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part. Any material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations.

3.3.4 Painting. Unless otherwise specified (see 6.2), the power source shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.5 corrosion protection. Corrosion protection of all metal components shall conform to the requirements of NEMA EW 1. The power source shall be designed and equipped for operation and storage outdoors in a humid, salt and dust laden shipyard environment, including adverse weather conditions and wind driven rain. All external and internal surfaces, internal components, wiring, and connections shall be protected to withstand extreme environmental conditions.

3.3.6 Portability and rough handling. Unless otherwise specified (see 6.2), the power source shall be base mounted. The power source shall be provided with means for lifting by crane or hoisting mechanism equipped with a hook, shackle, or clevis or by fork lift truck. The lifting device shall be firmly and rigidly attached to the power source, capable of supporting 2-1/2 times the power source's weight. Rough handling shall not cause distortion of the frame or any of its components.

3.3.7 Threads. All threaded parts used on the power source and its related attachment and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4 Power source components and requirements.

3.4.1 Voltage rating and regulation. The output voltage of the welder shall not exceed 87 volts at no load, shall be 80 volts $\pm$ 3 volts at 10 percent load, and shall be not less than 73 volts at rated full load. At no point shall the load voltage exceed the no load voltage. Primary line voltage compensation taps shall be provided on each power source. A minimum of 1 inch clearance shall be maintained from ground to taps and between taps. Identification of taps shall be permanently located on the tap panel. The taps shall not ground to frame in event of terminal board failure. Links shall be provided for changing tap connections.

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3.4.2 Current rating. The power source shall be rated at 100 percent duty cycle in accordance with NEMA EW 1 for class I machines (NEMA Class I(100)). The power source shall be capable of operating under 100 percent load in an maximum ambient temperature of not less than 110°F. The power source shall be capable of delivering 50 Percent overload for 2 minutes following its normal operation at 100 percent duty cycle in 110°F ambient temperature.

3.4.3 Polarity. The size 500 and 750 ampere power sources shall be of single polarity design. When specified (see 6.2), the size 1000 and 1500 ampere power sources shall be of dual polarity design. The power sources shall provide 1000 and 1500 ampere output, respectively, when operated in the single polarity mode, and shall provide not less than 500 ampere output in each polarity when operated in the dual polarity mode.

3.4.4 Voltmeter and ammeter. A DC, analog or digital voltmeter and ammeter shall be provided to measure the welding power output of the power source. The instruments shall be weather-tight, located on the front of the unit. The instruments shall be at least 2-1/2 inches in height and at least 3 percent accurate at full scale.

3.4.5 Pilot light. An "on-off" pilot light with a protective cover shall be provided on the front of the unit panel with the voltmeter and ammeter. The protective cover shall not protrude over 1 inch from the face of the enclosure. The light shall be visible in daylight from a distance of 10 feet.

3.4.6 Automatic overload protection. The primary line voltage feeding the main transformer and main control circuits of the power source shall be automatically removed in event of an overload. All protective devices shall be readily accessible for replacement or resetting.

3.4.7 Enclosure. The power source components shall be securely mounted within a rigid, all weather, metal enclosure conforming to the requirements of NEMA EW 1. The ventilation openings shall permit positioning multiple power sources adjacent to each other. Access to the primary and secondary terminals, transformer taps, and diode bridges shall be provided. Access openings shall easily removable for inspection, cleaning, and servicing. After wiring the unit, all internal components and connections shall be sprayed with air drying epoxy or varnish or two-part urethane or similar q aterial and dried according to the supplier’s recommended drying practice. The terminal board front with the links in standard positions shall be sprayed with a suitable material to prevent corrosion.

3.4.8 Heaters. When specified (see 6.2), heaters shall be mounted inside the power enclosure with the necessary controls to allow heating only when the power source is “off” at the front panel and the primary power is on to minimize moisture inside the enclosure. Heaters may be furnished when not specified provided the heater system is part of the manufacturer’s standard design.

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3.4.9 Ventilation system. Unless otherwise specified (see 6.2), a forced air cooling system shall be provided. When furnished, forced air cooling shall be provided by a single phase, totally enclosed, prelubricated, ball bearing fan motor(s). The fan shall provide ventilation and cooling of all component parts to meet the 100 percent duty cycle in 110°F ambient temperature. The fan assembly shall be mounted to withstand the vibration. Fan blades shall be of material and design that will prevent fatigue failure under conditions anticipated for these units.

3.4.9.1 Cooling fan control. When specified (see 6.2), the cooling fan shall be provided with a control device to operate the fan only when the transformer-rectifier system requires forced air cooling. Fan control may be furnished when not specified, provided the control system is part of the manufacturer's standard design.

3.4.10 Insulation. Transformer insulation shall conform to NEMA EW 1. The class of insulation system used shall permit continuous transformer operation and performance.

3.4.11 Temperature rise. The maximum allowable temperature rise of the various components of the power source shall conform to NEMA EW 1. Means shall be provided to remove the main transformer from the "primary power line before the temperature within the power source enclosure reaches a damaging temperature.

3.4.12 Transformer. The transformer shall produce a continuous current which, when rectified, shall have a direct current rating as specified in 1.2. Transformer assemblies shall either be dipped and baked in class F or better varnish and then mounted to the base also sealed with a material to withstand extreme environmental conditions or all transformer and diode assemblies shall be mounted to the base then dipped in air drying varnish and dried according to the supplier's recommended drying practice.

3.4.13 Diodes. In combining diodes to obtain required output current, no diode shall carry current in excess of its safe operating current under any steady state or transient condition of welder operation. Paralleled diodes shall share the current load (by using resistors, matched pairing, or other means). Surge suppression protection shall be provided with capacity to reduce any transient voltage peaks which would cause failure of the diodes from either AC or DC reflected transients. The diode bank or stack shall be removable as a unit to permit removal and replacement of individual diodes. The diode stacks shall be installed to minimize the collection of foreign particles in and around the stacks. All heat sinks shall be coated with an anti-corrosion material to withstand a salt water environment. All main power diodes shall be protected from environmental damage by a material which completely seals the device.

3.4.14 Electrical requirements. The power source shall be wired to conform to NFPA 70. Unless otherwise specified (see 6.2), the power source shall operate on 3 phase, 60 Hertz (Hz) input power. The primary Input voltage on the size 500 and 750 ampere power sources shall be either 230 volts or 460 volts, as specified (see 6.2). The primary input voltage on the size 1000 and 1500 ampere power sources shall be 460 volts.

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3.4.14.1 Primary connections. Access for the primary power leads shall be provided through the rear or side panel. All primary leads shall be connected inside the enclosure. The primary leads shall be attached conforming to electrical codes current on the date of the contract.

3.4.14.2 Secondary connections. Specified connectors shall be provided and installed on the secondary (output) bus bars (see 6.2). A minimum of four output circuits (four negative and four positive connections) shall be provided on each power source. The connectors shall be located within the enclosure, properly covered when welding cables are connected to the bus bars to prevent electrical hazard and the entrance of rodents. Space shall be provided between lugs and between the secondary bus bars and the enclosure to prevent forced bending of cables when attached to the bus bars.

3.4.14.3 Ground connections. Each power source shall have ground connections for both individual unit grounding and grounding to adjacent units when connected in parallel (see 3.4.15).

3.4.15 Parallel operation. When specified (see 6.2), the power source shall be designed to operate in parallel with two or more similar power sources of the same manufacture. When operated in parallel, the load shall be divided so that no individual power source will draw in excess of ten percent more of the load than any other individual power source operating in parallel with it. Means to isolate any one of the group of parallel power sources to protect against an internal fault or a short circuit of one of the other power sources shall be provided on each of the secondary bus connections.

3.5 Standard equipment. All standard equipment normally provided with the manufacturer's commercial power source shall be furnished.

3.6 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.7 Repair parts. Repair parts shall be furnished as specified (see 6.2).

3.8 Marking on instruments, diagrams, and plates. All words on instruments, diagrams, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.8.1 Nameplate. A nameplate shall be securely attached to each power source. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

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3.8.2 Diagram of connections. A diagram of electrical connections shall be attached to the inside of each power source. The diagram shall be protected with transparent varnish, transparent plastic covering, or other transparent material.

3.9 Workmanship. Workmanship of the power source shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor Is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Gov-ernment. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The rough handling test described in 4.7.3.9 shall be performed prior to performing any other test. The power source shall pass the first article examination and all tests to be accepted.

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4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.2 through 4.7.3.3, and the inspection in 4.8. The power source shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each power source shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit and 20 percent to 95 percent relative humidity.

4.7.2 Operational test. All circuits in the power source shall be energized. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified without performing any actual welding.

4.7.3 Performance test. Each power source shall be capable of performing all tests specified herein. Where applicable, tests shall follow the procedures of NEMA EW 1. At the option of the procuring agency, each power source shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2). All performance tests shall be completed without performing any maintenance on the power source.

4.7.3.1 Open circuit voltage test. The power source shall be tested to determine that the open circuit voltage meets the requirements given in 3.4.1.

4.7.3.2 Maximum current output test. The power source shall be tested to determine that the maximum current output meets the requirements for the maximum rated current output for the power source given in 3.4.2.

4.7.3.3 Dielectric tests. Dielectric tests shall be in accordance with NEMA EW 1.

4.7.3.4 Voltage regulation test. Load tests shall be conducted, showing conformance with voltage regulation when supply voltage is connected to applicable taps (see 3.4.1).

4.7.3.5 Overload tests. With the equipment operating in a thermally stabilized condition at rated load current, a 200-percent-of-rated-load current shall be applied. me power source shall comply with the requirements of 3.4.6.

4.7.3.6 Temperature test. Temperature tests shall be conducted in accordance with NEMA EW 1. Temperatures shall conform to requirements in 3.4.11.

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4,7.3.7 Thermostatic protection test. With the equipment operating in a thermally stabilized condition at rated load current and with enclosure panels in place, the fan motor shall be disconnected to determine that the thermostatic protection device meets the requirements of 3.4.11.

4.7.3.8 Parallel operation test. When the power source is designed to be operated in parallel with other similar power sources (see 3.4.15), the power source shall be tested for parallel operation. The unit to be tested and a similar unit shall be connected in parallel. The load current shall be adjusted from no-load to combined full load ratings in five equal steps and returned to no-load. For each load adjustment, the line voltage and output current for each power source shall be recorded. Neither power source shall indicate a load more than 10 percent greater than the load of the other power source at any setting, as required in 3.4.15.

4.7.3.9 Rough handling test. Each power source shall be subjected to jerk and drop tests and meet the test conformance requirements in accordance with NEMA EW 1.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements shall be in accordance with MIL-W-45562.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The welding power sources are intended for use with multiple-operator welding systems. The power sources shall furnish direct current, constant voltage for welding in their specified current range.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b, Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of Individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Additional safety and health requirements, if required (see 3.2.4).

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f. Painting, if different (see 3.3.4).

g. Portability, if different (see 3.3.6).

h. Dual polarity on 1000 or 1500 ampere machine, if required (see 3.4.3).

i. Heaters, if required (see 3.4.8).

j. Forced air cooling, if not required (see 3.4.9).

k. Cooling fan control, if required (see 3.4.9.1).

l. Input power, if different (see 3.4.14).

m. Input voltage on size 500 or 750 volt power source (see 3.4.14)

n. Specify required secondary connections (see 3.4.14.2).

o. Specify parallel operation, if required (see 3.4.15).

p. Specify and fully describe optional equipment, if required (see 3.6).

q. Specify applicable repair parts, if required (see 3.7).

r. Nameplate, if different (see 3.8.1).

s. Inspection conditions, if different (see 4.3).

t. First article inspection, if different (see 4.4).

u. Quality conformance inspection, if different (see 4.5).

v. Performance tests, if different (see 4.7.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
•	•	•	•
•	•	•	•

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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD From 1423.

6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the contract.

6.5 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a power source will operate. It does not limit the hazard level of individual power sources in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to Integrate this new power source into its future environment.

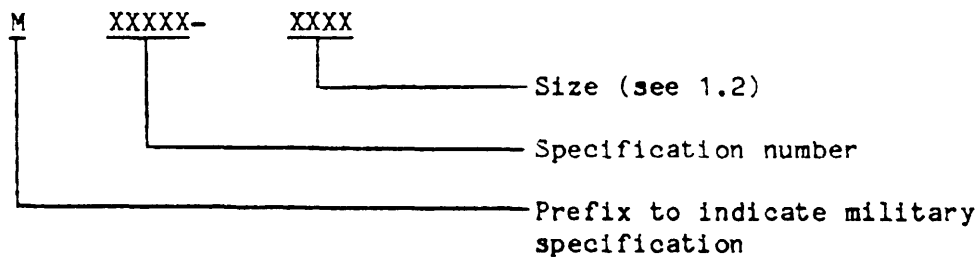
6.7 Training. Training required by the procuring activity should be provided as specified in the contract,

6.8 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.9 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

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6.10 Part or Identifying Number (PIN). The PIN to be used for power sources acquired to this specification are created as follows:



6.11 Subject term (keyword) listing.

Arc
Direct current (DC)
Multi-operator
Machine, welding
Transformer-rectifier
Voltage, constant
Welding machine

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - ME
Navy - As, YD
Air Force - 84

User activities:
Navy - CG, MC, OS, YD

Preparing activity:
DLA - IP

Project (3431-0195)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80040E	Welding Machine, Arc, DC, Trans-Rect, Constant Voltage, Multi

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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